

CS4800: Designing Sustainable ICT Systems

Lecture 1: Introduction

September 3, 2026

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Course team

- [Przemysław Pawełczak](#) (course admin and main lecturer)
 - Associate Professor at TU Delft leading <http://github.com/tudssl>
- TA team: Tomas Šutavičius
- Contact: CS4800-EWI@tudelft.nl (you can try your luck with my personal email, but the odds are against you)

Course made possible by [SIDN Fund](#) support through grant "[Educational ecosystem aimed at sustainable ICT system design](#)"

Learning objectives

1. **List** the methods for assessing ICT industry climate impacts
2. **Comprehend** the implications of different ICT system design decisions on the climate
3. **Apply** ICT system design techniques that minimize climate impact
4. **Analyze** how each ICT system design technique affects climate impact
5. **Create** new methods of ICT system design with sustainability as a core performance metric

Course organization

Lectures

- Thursdays and Fridays, 15:45-17:30, on campus (check MyTimetable!)
- Each cover one component of sustainable ICT
- Course material: lecture slides + lecture notes (posted after each lecture)

Labs

- Thursdays, 15:45-17:30 (check MyTimetable!)
- Project work: presentation and feedback

Detailed information in official [course repository](#); announcements on Brightspace

Course organization

Lecture 1: Introduction

Lecture 2: Defining Sustainability

Lecture 3: Sustainable Software

Lecture 4: Sustainable AI

Lecture 5: Sustainable Cloud Computing

Lecture 6: Sustainable Electronics Design

Lecture 7: Sustainable Data Storage

Grading and assessment

70% practical assignment

- Two possible assignments (sustainable software/AI), described later today
- Work in groups of two/three
- Meant to develop deeper understanding of specific topic
- 60% report grade, 10% presentation grade

30% exam

- Multiple choice, single answer questions, practice questions available at end of each lecture
- Based on lecture slides + lecture notes
- Meant to test course knowledge
- Wednesday, November 4, 13:30-15:30, AE Hall E / Wednesday, January 20, 13:30-15:30, ME Hall H (resit)

AI policy

You are allowed to use LLM tools for:

1. Coding
2. Writing - **BUT** you are responsible for everything you write! Any claims you make must be based on verifiable technical/scientific sources!
3. Conceptualization and ideas - **BUT** if you do, you must upload a complete record of your interactions in an appendix to the assignment

What will you learn in this lecture?

Lecture 1: Introduction

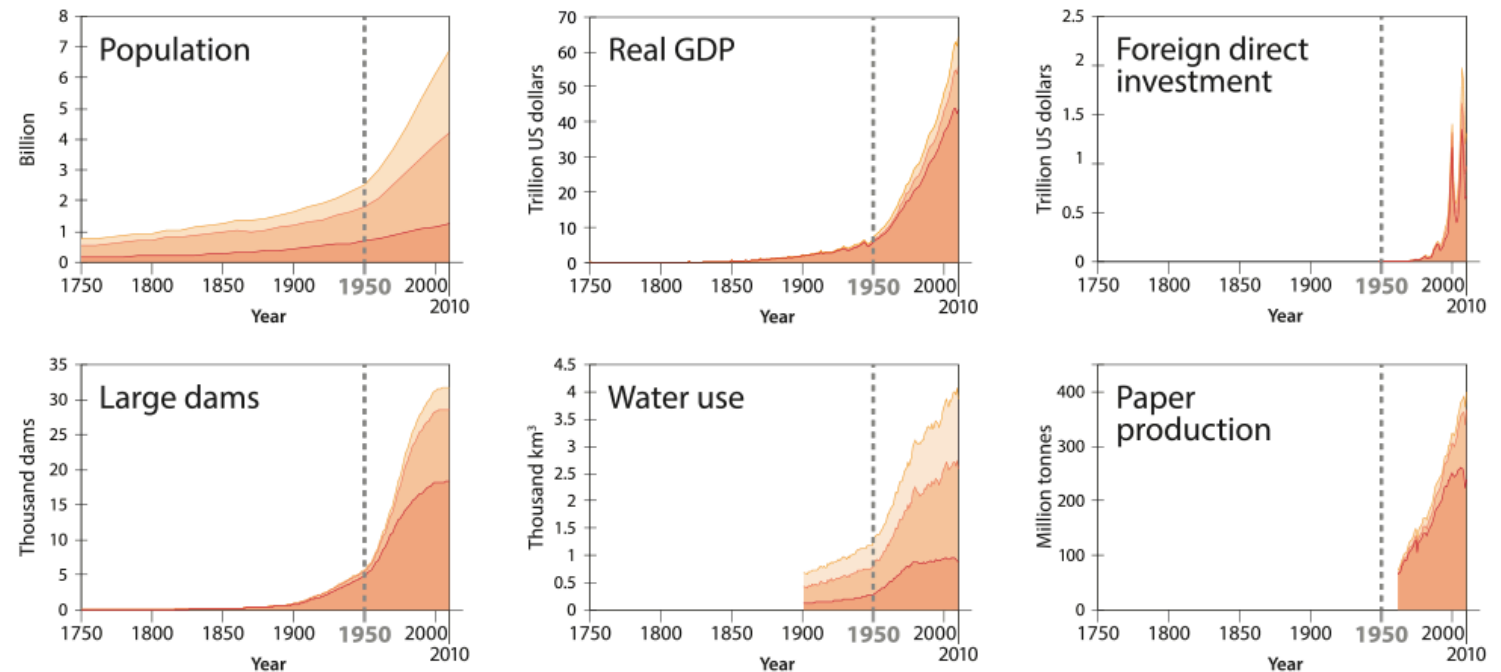
- Basics of climate change
- ICT impacts on the environment
- Prospective solutions
- Economic aspects

"The Great Acceleration"

- Human society has been [expanding](#) continuously and exponentially
- Human population, economic development, and technological innovation have grown

Socio-economic trends

OECD BRICS Others

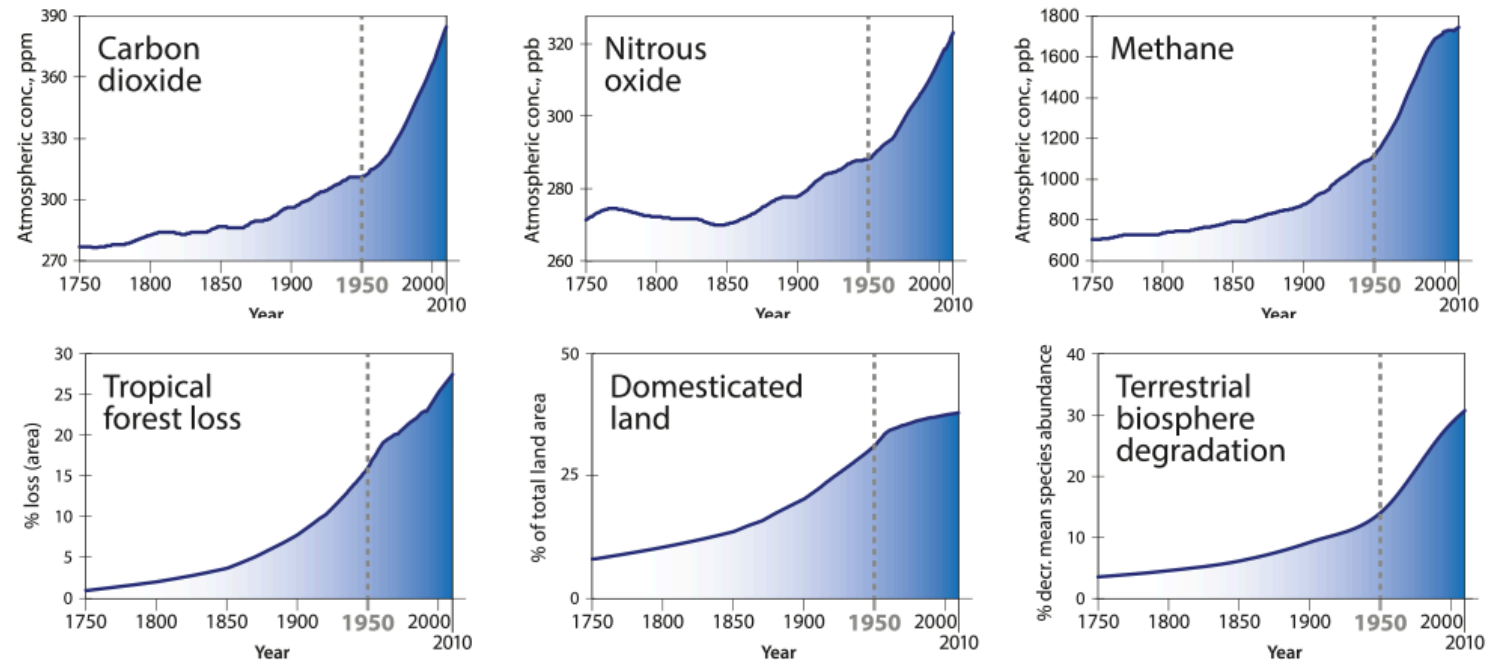


[Historic socio-economic trends](#)

"The Great Acceleration"

- ... but so have environmental impacts of human society on Earth system!

Earth system trends

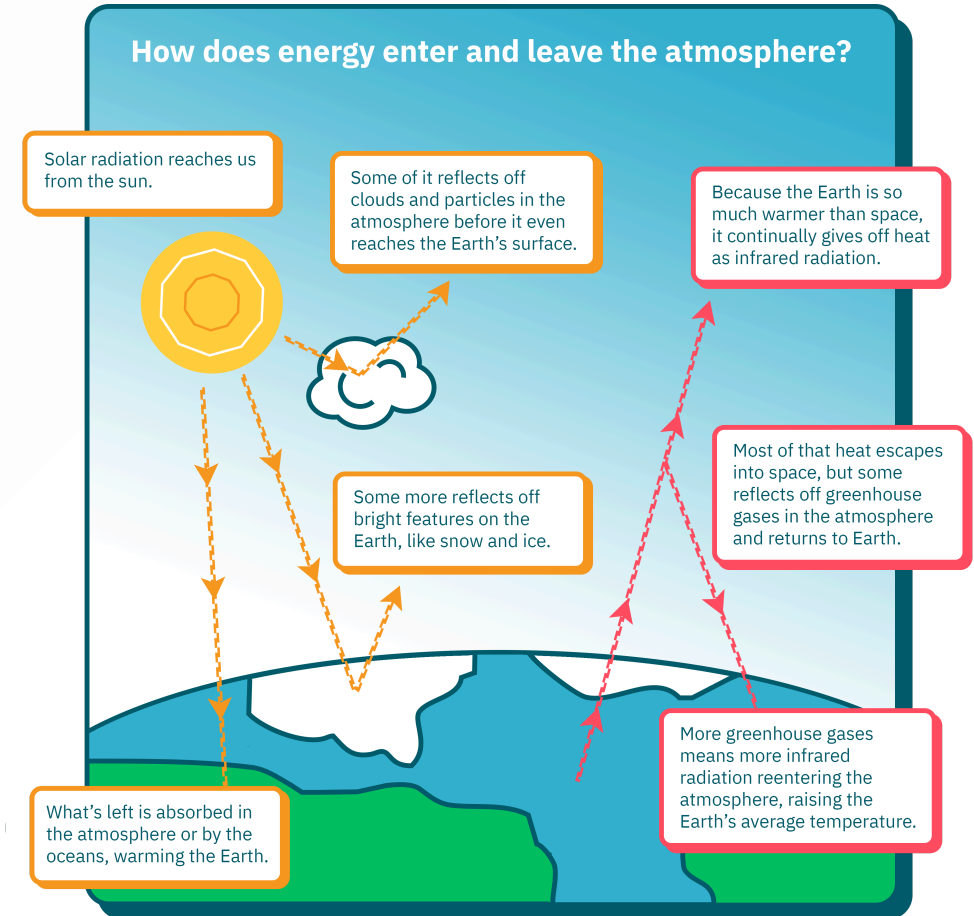


Historic Earth system trends

Climate change

- Earth is in energy equilibrium: incoming solar radiation is equal to infrared energy emitted
- Radiated energy is correlated with Earth's temperature:

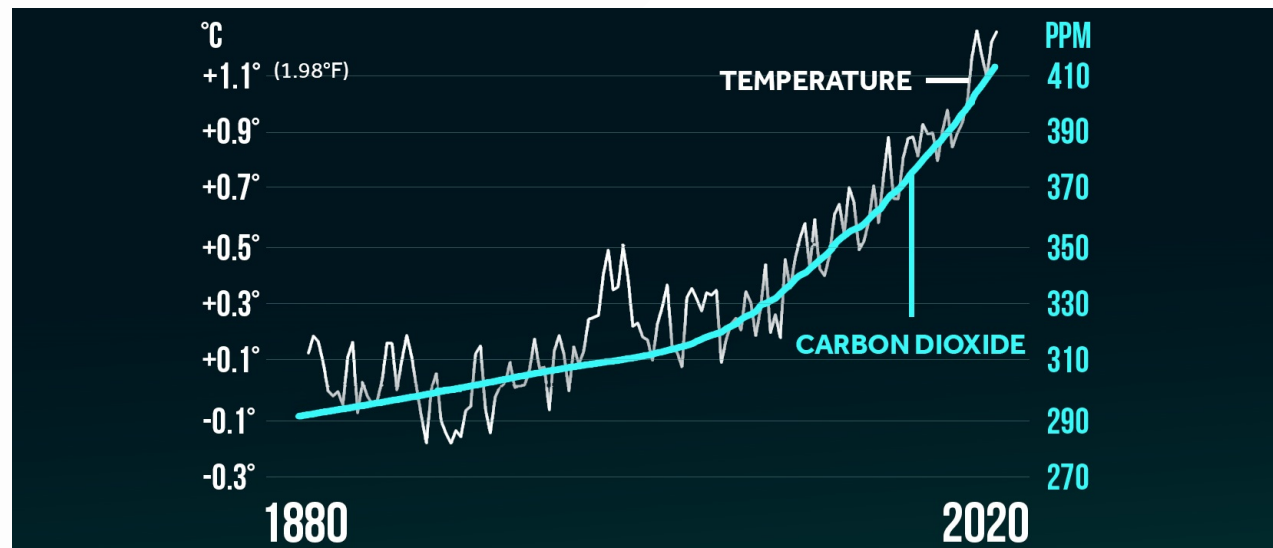
$$\text{infrared radiation} \propto T^4$$



The energy balance within Earth's atmosphere

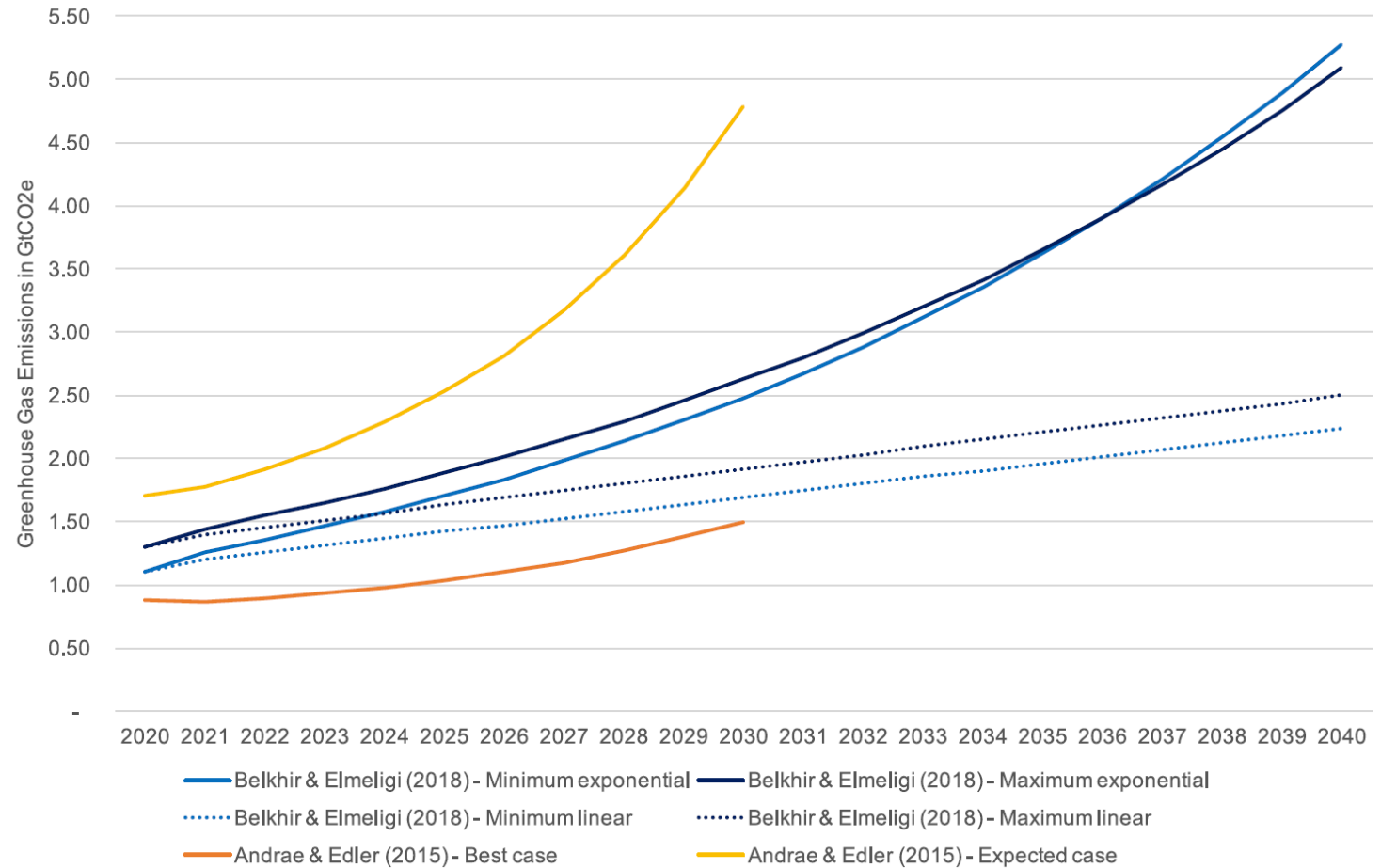
Climate change: greenhouse gases

- World economy relies on combustion of fossil fuels
- [Greenhouse gases](#) (GHG's) like CO₂, methane are introduced into atmosphere
- GHG's absorb infrared energy emitted by Earth
- Atmosphere heats up to maintain equilibrium → global warming!
- Warmer atmosphere also stores more water → extreme weather patterns!



ICT emissions

- Software requires hardware to run it
- Hardware requires energy to run
- Energy production emits GHG's
- Therefore, ICT growth means GHG emissions increase!
- 2020: up to 4% of global emissions, greater than aviation, maritime sector

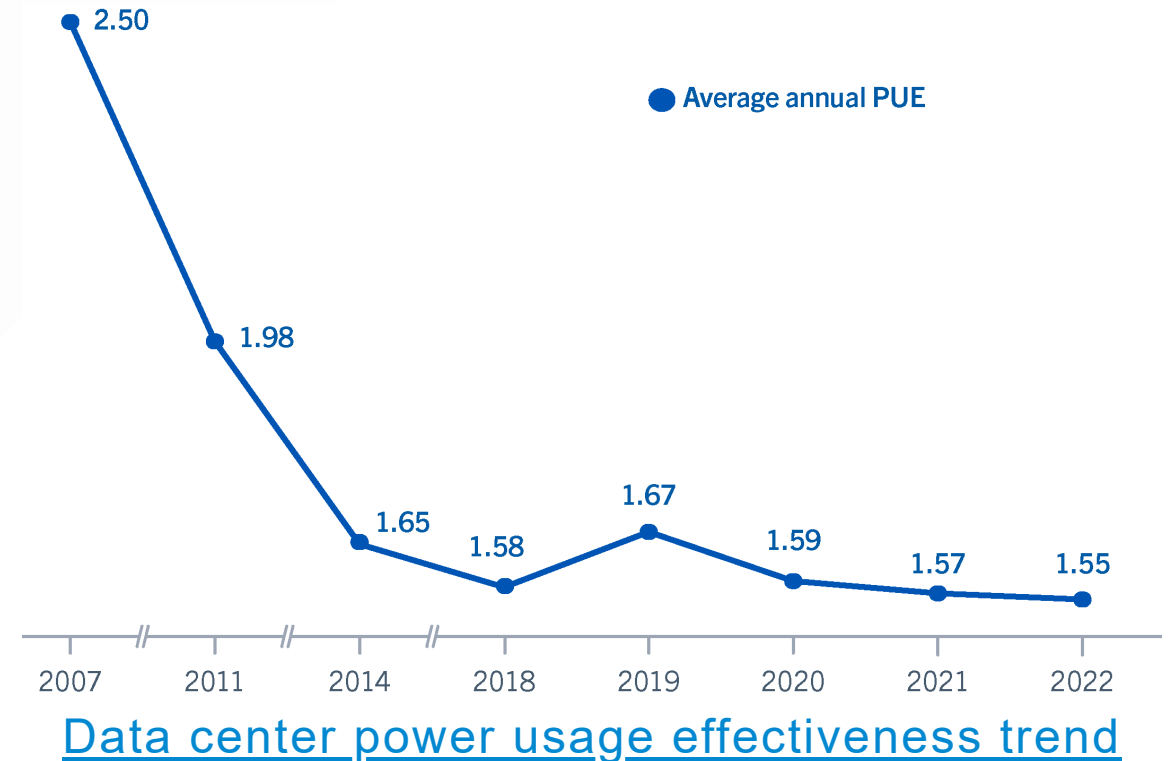


Various estimates for ICT emissions growth

What solutions have been proposed to this problem so far?

Prospective solutions: improving efficiency

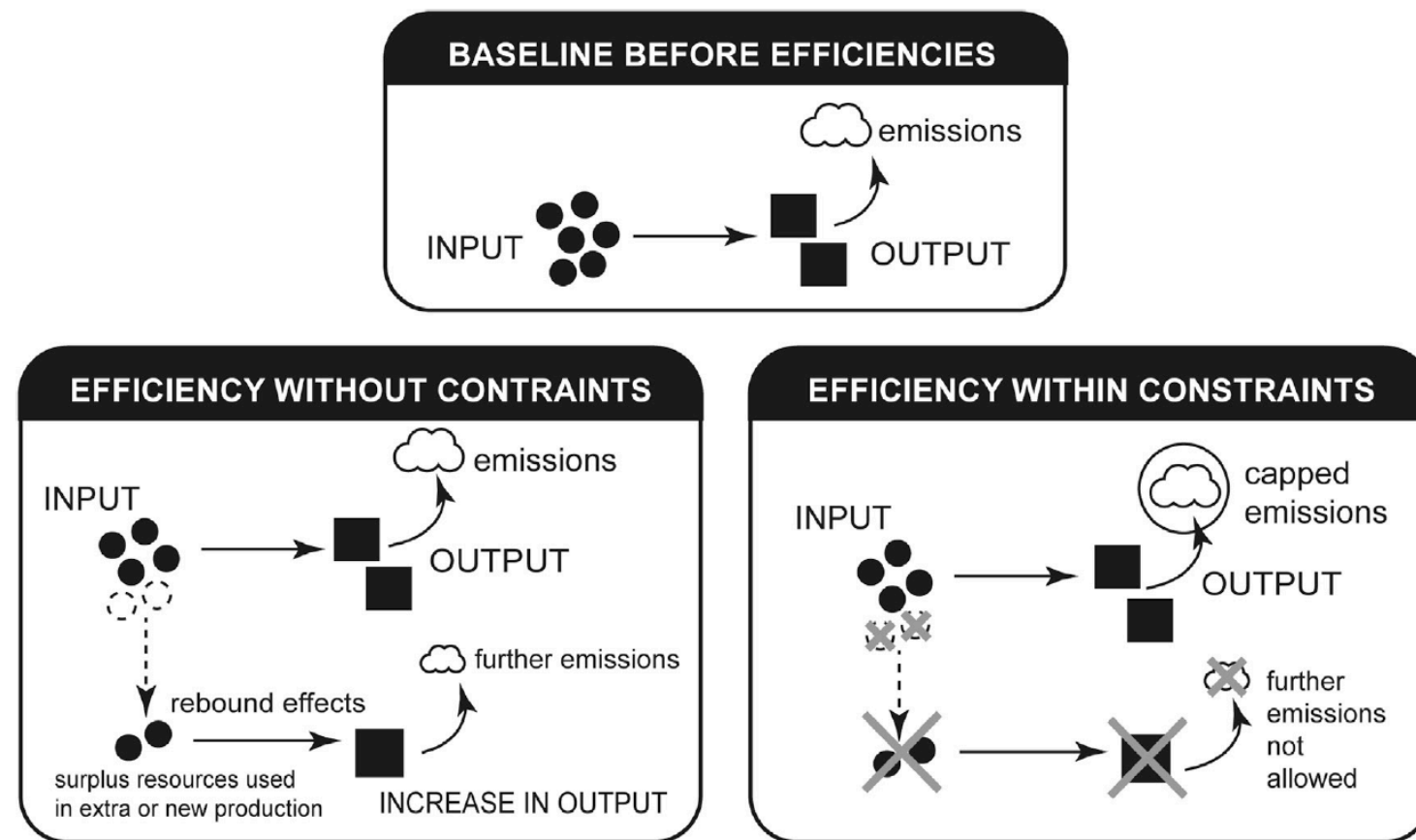
- More efficient ICT systems consume less energy per unit of compute
- Historically: data center workloads increased by 550%, emissions kept stable
- Moore's Law stalling, thermodynamic limits ('quantum entanglement') near
- Efficiency gains slowing



Prospective solutions: rebound effects

1. More efficient process requires less input per output...
2. Cost per output becomes lower...
3. ... so more people use it!

→ Efficiency improvements eaten up by [rebound effects](#), unless specifically constrained by regulation



[Limiting emissions only possible by constraining further consumption from rebound effects](#)

Prospective solutions: saturation

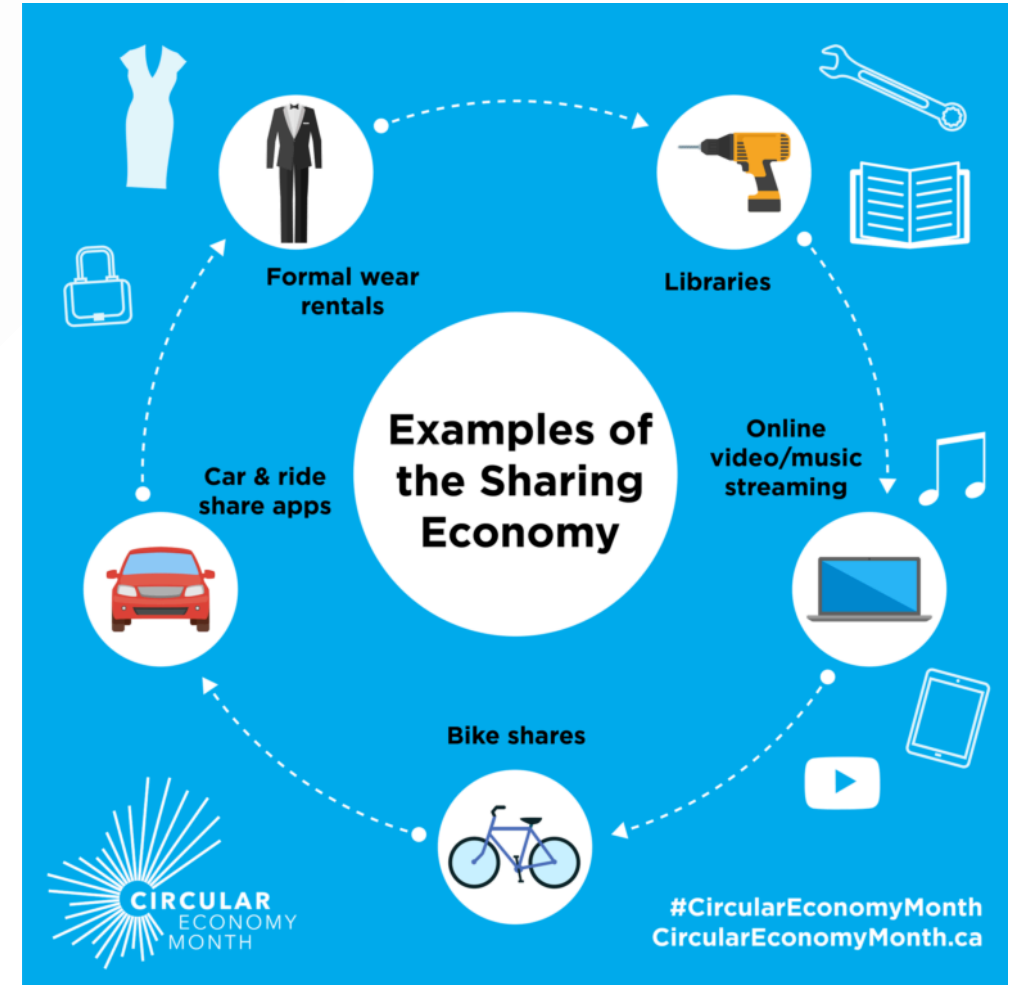
- Once every person on Earth has a smartphone, we should reach a [theoretical cap](#) for emissions
- However, companies keep releasing new products and services
- Saturation is incompatible with how the economy works!



[Planned obsolescence makes saturation unlikely](#)

Prospective solutions: digital sharing

- Industry [claims](#) that ICT can help decarbonize other sectors
- Such effects have not yet [materialized](#)
- The [digital sharing economy](#) might partially help: people use products only when they need them, reducing the amount of products required



[Examples of digital sharing.](#)

Prospective solutions: carbon removal

- Some form of [carbon removal](#) ([wetland restoration](#), [ocean capture](#), [enhanced rock weathering](#)) will probably be necessary for 'hard-to-abate' sectors
- Tree planting and other natural methods are [insufficient](#)!
- (Still nice to do it)



[Planting a few trees can't hurt](#)

Example: carbon capture and storage

- Industrial [carbon capture and storage](#) (CCS) or utilization (CCU): scrubbing of industrial exhaust gas
- Exhaust gas is chemically treated at high temperatures → high energy consumption, needs special chemical solvents
- Resulting CO₂ stream stored or used as feedstock



[Industrial CCS facility in Japan](#)

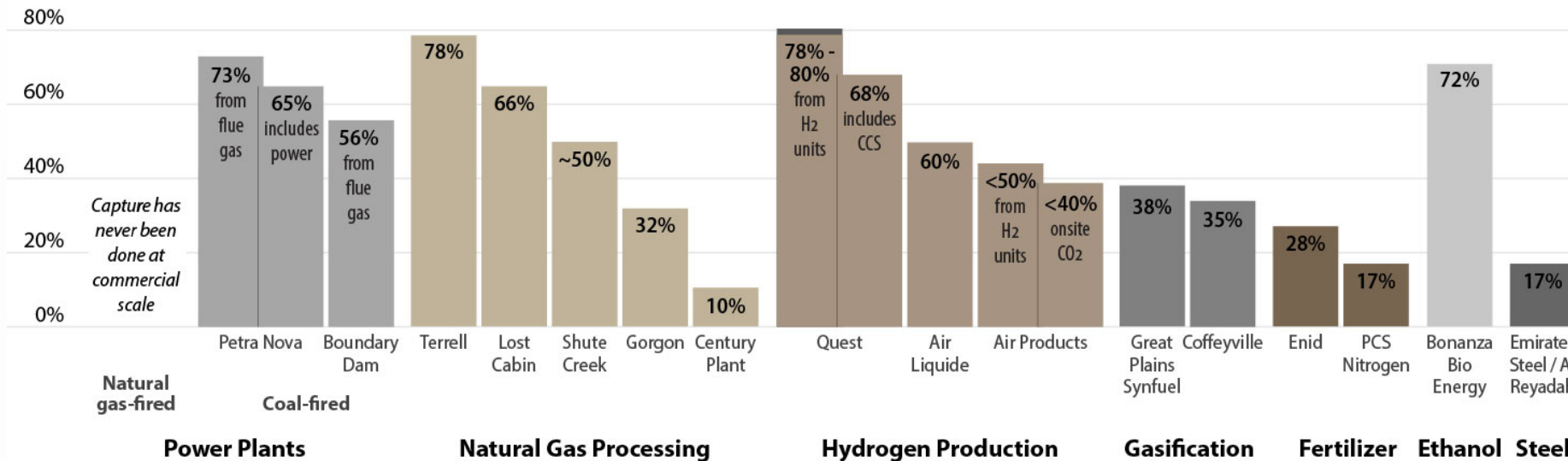
Example: carbon capture and storage

- CCS technology [does not work](#) in real life as well as proponents claim
- Requires additional energy, which [could be used to directly decarbonize](#)
- We would need [enormous amounts](#) of chemicals and energy to capture all global emissions

Real-World CO₂ Capture

100% carbon capture

95% or higher: Industry claims for CO₂ capture



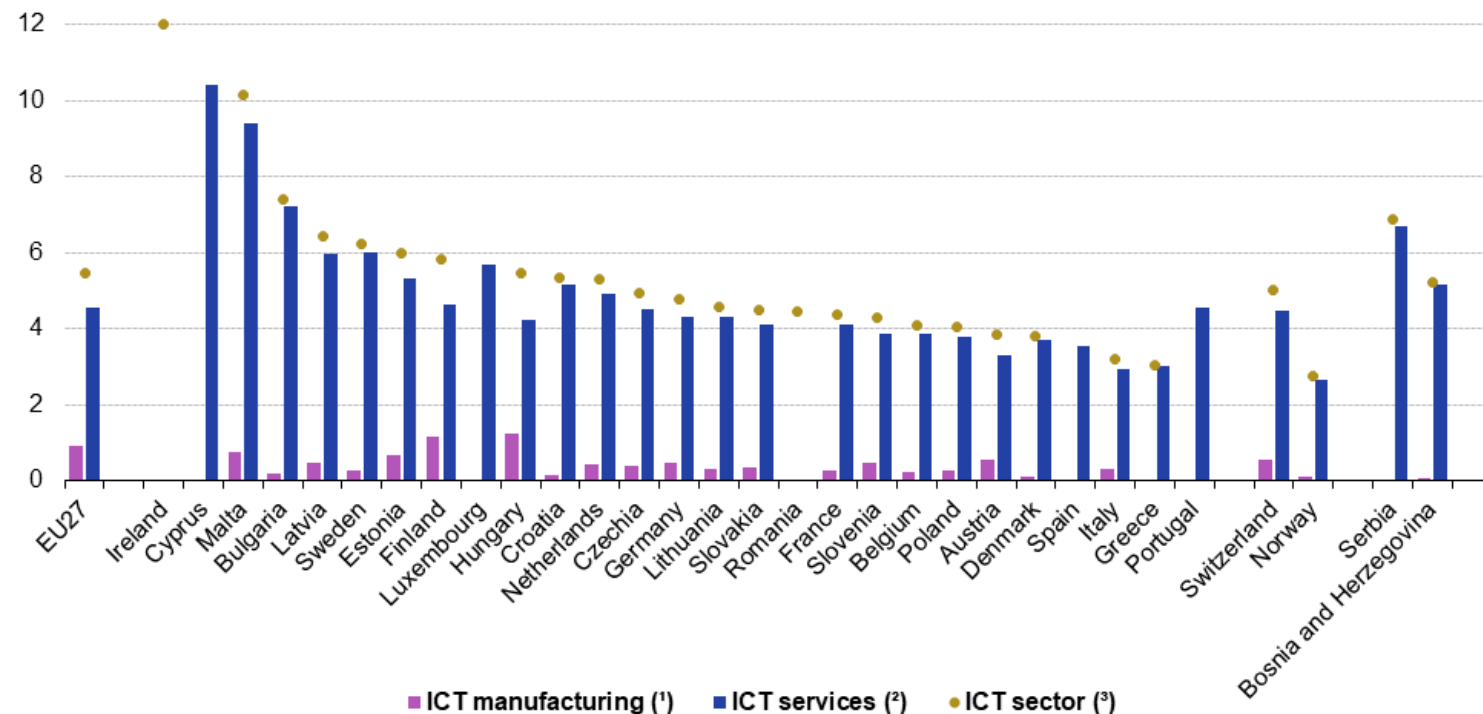
[Carbon capture rates in real-world projects vs industry claims](#)

How do we engage with the planet?

Economics: GDP

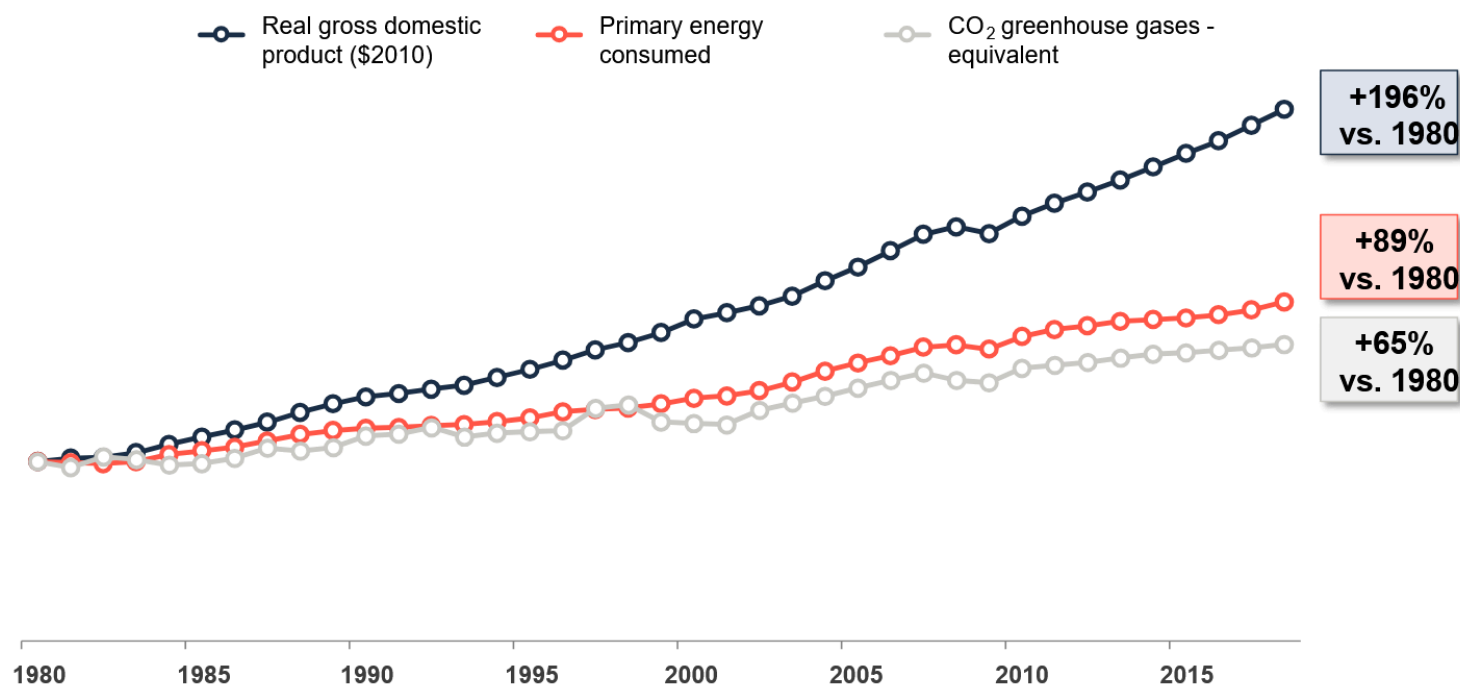
- Planet seen as natural capital, extraction produces economic value
- Economic activity measured through [Gross Domestic Product](#) (GDP)
- Example: software contribution measured with standard [System of National Accounts](#)
- Economic growth traditionally seen as [necessary](#)

Value added for the ICT sector, EU, 2022
(in %, relative to gross value added)



Economics: energy and emissions coupling

- All economic activity requires energy: [energy-GDP coupling](#).
- Energy currently comes from fossil fuels, so GDP and emissions are also [coupled](#)
- No sustained absolute [decoupling](#) has been observed anywhere!



[Relative decoupling between GDP growth, energy consumption, and emissions](#)

Economics: growth or degrowth?

- Is it even [possible](#) to decouple growth, energy consumption, and emissions?
- Complex economic, political, and philosophical question -> lots of opinions, (respectful) debate invited
- In this course: focus on reducing environmental impacts of ICT systems!
 - Improve software efficiency
 - Study data center design
 - Trade-off costs and benefits of AI
 - Redesign electronics

Recap

- Climate change is caused by combustion of greenhouse gases
- The ICT sector causes a significant part of global emissions
- Proposed solutions are important, but not sufficient to decrease impacts
- Significant future questions about GDP growth and coupling

Exam(ple) questions

1. **Why is carbon capture insufficient to compensate for global emissions? Choose the false answer.**
 - a. Practical applications have capture rates below what we would desire
 - b. There isn't enough space underground to store all the carbon dioxide we capture
 - c. It would take up a huge amount of materials to capture and transport all emissions
 - d. It would be more efficient to use the energy to directly decarbonize
2. **Why are critics questioning the possibility of GDP-emissions decoupling?**
 - a. It is impossible to fully eliminate the use of fossil fuels
 - b. It is impossible to support human well-being without economic growth
 - c. No sustained absolute decoupling has been observed anywhere in the world
 - d. It is impossible to convince people to stop consuming carbon-intensive products

Exam(ple) questions

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Feedback

- We need your feedback to continue improving the course!
- Feedback on Lecture 1 slides/notes:



Project Organization

- Two possible assignments: sustainable software or sustainable AI
- Assignment specification posted on Brightspace
- Deliverables: project report (60% of course grade), final presentation (10% of course grade)
- Make your own groups of 2/3, choose preferred assignment
- Diversity of perspectives preferred (don't just pick your friends!)
- Sign up on Brightspace by **Thursday, September 10, 15:45** - mandatory to participate in course!
- Midterm presentation (formative) on **Thursday, October 8, 15:45**
- Final presentation (graded) on **Thursday, October 22, 15:45**

Assignment: Sustainable Software

- Simulating typical server workloads to assess energy consumption
- Implementing just-in-time (JIT) compilation to improve energy efficiency
- Assessing environmental impact: carbon emissions
- Includes coding tasks: Linux, Java Spring Boot, Python

Assignment: Sustainable AI

- Designing AI system for conservation case study: data, training strategy, deployment strategy
- No model training required: focus on system design
- Assessing environmental impact: carbon emissions, water consumption, material use

Thank you for your attention!

Next lecture: Defining Sustainability

Thursday, September 10, 15:45-17:30, EEMCS Hall F

- Conceptualizing sustainability
- Defining sustainable ICT systems
- Frameworks for sustainability: planetary boundaries, circular economy
- Sustainability indicators: global warming potential, emissions reporting, carbon/energy/material/water intensity
- Social impacts of ICT

Questions?